INCH-POUND

MIL-DTL-46625D
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SUPERSEDING
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DETAIL SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, M3A2, ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M3A2 Cartridge Actuated Initiator (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-2550 Ammunition Genera Specification For
MIL-I-14802 Initiator, Cartridge Actuated, M3A2, Remanufactured Parts Assembly
MIL-DTL-46559 Initiator, Cartridge Actuated, M3A2, Metal Parts Assembly

Comments, suggestions, or questions on this document should be addressed to DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E12AP, Document Control, 4123 Artisans Court, Suite 103, Indian Head, MD 20640-5115 OFFICIAL BUSINESS, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 1377

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MS28741 Hose Assembly, Detachable End Fitting, Medium Pressure

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CAGE Code 19200 and 19203)

DL8595821 Initiator, Cartridge Actuated, M3A2 Assembly
8594157 Cartridge, Impulse, M73 Assembly
8796541 Box, Fiberboard (Inner) Packing, Ammunition, for Initiator, Cartridge Actuated, M3A2 or M5A2
8796542 Box, Packing, Ammunition, for Initiator, Cartridge Actuated, M3A2 or M5A2
9216934 Box, Fiberboard (Outer) Alternative Packing, Ammunition, for Initiator, Cartridge Actuated, M3A2 or M5A2

(Application for copies should be addressed to the DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E11G7, Document Control, 4072 North Jackson Road Suite 106, Indian Head, MD 20640-5115 OFFICIAL BUSINESS.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

American National Standards Institute (ANSI)/American Society For Quality (ASQ)

ANSI/ASQ Z1.4 Sampling Procedures and Tables for Inspection by Attributes (DoD adopted)

ANSI/ISO/ASQ A8402 Quality Management and Quality Assurance Vocabulary

(Copies of this document are available online at http://www.asq.org/, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

American Society For Testing And Materials (ASTM)

ASTM E1742/E1742M Standard Practice for Radiographic Examination

(Copies of this document are available online at http://www.astm.org/ or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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International Organization for Standardization

ISO 3601-3, ―Fluid power systems — O-ring — Part 3: Quality acceptance criteria

(Copies of this document may be purchased online at http://www.iso.org/)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. GENERAL REQUIREMENTS.

3.1 Materials. All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure there from has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2 Primary components. For the purpose of this specification, the M73 impulse cartridge (Drawing 8594157) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of initiators. One primary component production lot may be used in more than one initiator production lot.

3.3 Firing pin release. The firing pin of the assembled initiator, less safety pin, cartridge, cartridge retainer, and chamber, shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin 3/4 inch maximum (max).

3.4 Initiator pin security. The initiator pin of the assembled initiator, less cartridge, cart cartridge retainer, and chamber, shall not withdraw beyond the safety pin when subjected to an axial load of 200 to 220 pounds. Application of the load shall not cause permanent deformation of the safety pin.

3.5 Breakaway torque. The initiator chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch-pounds without loosening or deformation (see 4.2.3.1).

3.6 Residual magnetism. The initiator shall not deflect the indicator of the compass more than five degrees in either direction using the procedures in 4.3.2.

3.7 Functional.

3.7.1 Ballistics. Initiators loaded with accepted cartridges and temperature conditioned shall perform within the pressure values and ignition delays specified in table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, assembly number MS28741-4.

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Table I. Pressure values.

Temperature (°F)	Pressure (psi)		Ignition delay, max, milliseconds (see 6.5.2)
	Minimum	Maximum	
-60 to -70	550	1450	65
65 to 75	700	1550	65
195 to 205	700	1700	65

3.7.2 Misfires. There shall be no misfires.

3.7.3 Mechanical failures. The initiator shall function without mechanical failure (see 6.5.1)

3.8 Workmanship. Initiators shall be free from cracks, splits, incomplete of damaged threads on inlet and outlet ends, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease rust, corrosion, or other foreign matter.

3.9 Radiographic examination. Each initiator assembly shall be radiographically examined for defects as specified in 4.4.

3.10 Component parts. All component parts shall meet the requirements of MIL-I-14802 or MIL-DTL-46559 prior to assembly.

4. VERIFICATION

4.1 Quality assurance terms and definitions. Reference shall be made to ANSI/ISO/ASQ A8402 to define the quality assurance terms used.

4.1.1 Inspection. Inspection shall be in accordance with MIL-A-2550 and ASTM E1742/E1742M.

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 68to 77 °F
b. Altitude: Sea level
c. Humidity: 50 percent relative, max
d. Atmospheric pressure: 14.7 psia.

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4.2.1 Lot.

4.2.1.1. Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with ANSI/ASQ Z1.4.

4.2.1.2 Size of lot. A lot shall consist of an identifiable group of assembled initiators containing metal parts assemblies produced by a homogeneous process and loaded with one lot of accepted cartridges.

4.2.2 Examination. One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4X) magnification may be used for inspection of such defects as burr and foreign matter.

4.2.2.1 Cartridge, Impulse M73, Drawing 8594157, covering a detail of Drawing 8595821 or 10524101.

Characteristic	Defect	Method of Inspection
Critical		
1	Mixed cartridges	Visual
2	Metal defective	Visual
3	Marking missing, incorrect, or illegible	Visual
Major		
101	Foreign Matter present	Visual
Minor		
201	Case scratched or dented	Visual

4.2.2.2 Assembly, Drawing 1052401 or 8595821, less Drawing 8595404, 8593013, 8593014, and 839536.

Characteristic	Defect	Method of Inspection
Critical		
1	Mixed initiators	Visual
2	Initiator not in cocked position	Visual
3	Safety pin missing or improperly engaged	Visual
4	O-ring missing	Visual
Major		
101	Thread damaged	Visual
102	Foreign matter present	Visual
Minor		
201	O-ring not lubricated	Visual

4.2.2.3 Assembly Drawing 10524101 or 8595821, less Drawing 8593014 and NAS832.

Characteristic	Defect	Method of Inspection
Critical		
1	Cartridge loose	Visual/Manual
2	Cartridge retainer loose	Visual/Manual
3	Metering groove of retainer missing	Visual
4	Elongated hole of cartridge retainer missing	Visual
Major		
None Defined		
Minor		
None defined		

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4.2.2.4 Assembly, Drawing 10524101 or Drawing 5895821.

Characteristic	Defect	Method of Inspection
Critical		
1	Location of cartridge retainer	Gage/x-ray
2	Any part missing	Visual/x-ray
3	Improper assembly	Visual/x-ray
4	Foreign matter in gas port	Visual
Major		
101	Length from cap boss face to gas port face, max	Gage
102	Height of cap, max	Gage
103	Cap set screw missing, loose or above flush	Visual
Minor		
201	Total length, max	Gage
202	Location of marking incorrect	Visual
203	Protective finish inadequate or defective	Visual
204	Protective cap missing	Visual

4.2.2.5 Box, Fiberboard (Inner) Packing Ammunition Drawing 8796541

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Gap between body and cover. max	Manual
202	Contents loose	Manual
203	Metal ends loose or distorted	Manual
204	Cut or damaged through all impregnated layers	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Tape incomplete, torn, or badly wrinkled	Visual
207	Length of tear tab, min	Visual
208	Color of tape incorrect	Visual

4.2.2.6 Box, Packing, Ammunition, Drawing 8796542

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly assembled	Manual/Visual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units contained incorrect	Visual

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4.2.2.7 Box, Fiberboard (Outer) Alternative Packing, Drawing 9216934.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.2.3 Inspection testing. The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque. Each assembly shall be subjected to the breakaway torque test of the threaded joint between the cap and the chamber. Failure of any assembly to comply with 3.5 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism. Each initiator shall be subjected to the test for excessive residual magnetism as specified in 4.3.2. Failure of any initiator to comply with 3.6 shall be cause for rejection of the initiator.

4.2.3.3 Firing pin release. The initiator pin of each subassembly shall be subjected to the firing-pin release test. Failure of any subassembly to comply with 3.3 shall be cause for rejection of the subassembly.

4.2.3.4 Initiator pin security. The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any subassembly to comply with 3.4 shall be cause for rejection of the subassembly.

4.2.3.5 Functional tests. The following tests shall be performed by the Government.

4.2.3.5.1 Ballistics. Twenty-one units shall be tested to determine compliance with 3.7.1. The units shall be temperature conditioned as provided in table II and 4.3.5.2. Failure of any unit to comply with 3.7.1 shall be cause for rejection of the lot. Ballistic test samples for small lots shall be:

Lot size	Ballistic Sample
Less than 9	4
9 – 25	6
26-50	8
51-90	10
91-150	14
Over 150	21

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Table II. Conditioning temperature.

Temperature (°F)	Sample size
-60 to -70	7
65 to 75	7
195 to 205	7

4.2.3.5.2 Retest. There shall be no retest.

4.3 Test methods and procedures.

4.3.1 Breakaway torque. A torque wrench set at the torque requirement specified in 3.5 shall be used in conjunction with a fixture to determine the adequacy of the joints. The chamber shall be secured in a suitable vise and the torque shall be applied to the cap

4.3.2 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in a north-south horizontal position (cap end south) with the compass set in an east-west heading. The assembly shall be moved at a speed not greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degrees rotation of the assembly for four full passes.

4.3.3 Firing pin release. The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

4.3.4 Initiator pin security. The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.3.5 Functional.

4.3.5.1 Ballistics. The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.3.5.2. Required pressure-time readings shall be taken as specified in 3.7.1.

4.3.5.2 Temperature conditioning. Each assembled initiator shall be conditioned for a period of time not less than 4 hours nor greater than 24 hours. When reconditioning initiators at -60 °F, all condensation shall be removed from the interior surfaces before returning the initiator to the conditioning chamber.

4.4 Radiographic examination. All assembled initiators shall be radiographically examined in accordance with ASTM E1742/E1742M, with exception that an assembled M3A2 initiator will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly

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perpendicular to the rays of the X-ray machine. All initators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. Any missing serial numbers are to be identified on the X-rays. The initiators shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon, The radiographic plate identification shall include item nomenclature, the complete lot number assigned to that lot, the span of serial numbers displayed, and the contract number. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective initiators found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. O-ring present	P/N MS28755-214
b. Steel balls present (3 ea)	P/N MS19060-3
c. Firing pin	P/N 8596168
d. Spring, initiator	P/N 8595406
e. Seal	P/N 10524102
f. Firing pin housing	P/N 8596169
g. Retainer cartridge	P/N 8596013
h. Cap set screws, No. 10 (.190)	
i. Cartridge, Impulse M73	P/N 8594157
j. Pin, initiator	P/N 8595403
k. Chamber	P/N 8593014
l. Cap	P/N 8595404

4.5 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Intended use. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems. Since the assembly was designed for use in military aircraft, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. If required, the specific issue of individual documents referenced (see 2.2)
c. If first article inspection is required (see 3.2 and 6.3)
d. The responsible inspection activity (see 4.1)
e. Level of marking and packing (see 5.1, 6.5.1)

6.3 Contract packaging, packing and marking. The following requirements should be specified in the contract.

6.3.1 Packing and marking should be level A or B, as specified by the contracting activity/

6.3.1.1 Level A should be in accordance with Drawing 8796541 and 9796542 or SPI TPO Foo-138-3770.

6.3.1.2 Level B should be in accordance with Drawing 9796541 and 9216934 or SPI TPO F00-138-3770.

6.3.1.3 Lot numbers. Lot numbers should be in accordance with MIL-STD-1168.

6.4 Definition.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5. Packaging.

6.5.1 Packing and marking. Packing and marking should be levels A or B as specified by the contracting activity:

Level A in accordance with Drawings 8796541 and 8796542
Level B in accordance with Drawings 8796541 and 9216934

6.5.2 Air Force packaging. Air Force packaging should be in accordance with TPO F00-138-3370.

6.6 Subject term (keyword) listing.
Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving system, aircrew

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes. The changes in this issue were authorized by ECP 09E21A010DBR1.

Custodian:
Navy-OS
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